

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)

21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

*I **P. Reddy Suhail(192312671L)** (Name / Reg No) certify that this submission is my original work and that*

I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Suhail

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

AI-ENABLED SMART SURVEILLANCE SYSTEMS OVER 5G NETWORKS

A Low-Latency Framework for Real-Time Intelligent Video Monitoring

Abstract— Surveillance systems are increasingly required to detect and respond to events in real time across large areas, but traditional camera networks generate huge volumes of video that overwhelm bandwidth and centralized processing. This paper explores how Artificial Intelligence (AI) combined with 5G connectivity can enable smart surveillance that is fast, accurate, and scalable. AI models such as convolutional neural networks perform object detection, facial recognition, and anomaly detection directly on video streams, while 5G's high bandwidth and ultra-low latency allow near-instant transmission of alerts and footage between cameras, edge nodes, and control centers. The proposed system processes video close to the camera using edge AI and uses 5G to coordinate across a distributed network of surveillance points. This approach can support faster threat detection, reduced bandwidth usage, and more reliable monitoring in smart cities, campuses, and critical infrastructure.

Keywords— *Artificial Intelligence, 5G, Smart Surveillance, Computer Vision, Edge Computing, Object Detection, IoT.*

I. INTRODUCTION

Public safety, campus security, and critical-infrastructure monitoring increasingly rely on dense camera networks. Traditional CCTV systems stream continuous video to a central server for human review, which is slow, labor-intensive, and does not scale as camera counts grow. Meanwhile, 5G networks offer high bandwidth and ultra-low latency, making it practical to move intelligence closer to the cameras and coordinate large numbers of them in real time. Combining Artificial Intelligence (AI) with 5G allows a surveillance system to detect, classify, and respond to events automatically, rather than depending on constant manual monitoring of video feeds.

II. PROPOSED SYSTEM

Camera / IoT Sensors → 5G Network → Edge
AI (Detection/Recognition) → Alert Engine
→ Control Center / Cloud

Cameras and IoT sensors capture continuous video and environmental data. This data is transmitted over the 5G network to nearby edge nodes, where AI models perform object detection, motion tracking, facial or license-plate recognition, and anomaly detection close to the source. When the models identify a relevant event — an intrusion, an unattended object, or unusual crowd behavior — the system raises an alert and forwards the relevant clip to a control center or the cloud for verification and record-keeping, rather than sending every frame from every camera.

III. AI-BASED SURVEILLANCE TECHNIQUES

Convolutional neural networks (CNNs) and modern object-detection models (e.g., YOLO-family architectures) identify and classify people, vehicles, and objects in each video frame. Recurrent and temporal models track movement across frames to flag loitering, unusual paths, or crowd-density spikes. Facial and license-plate recognition modules support identity verification at access points, while anomaly-detection models

trained on normal-activity patterns flag behavior that deviates from the baseline. Running these models at the edge, close to the camera, reduces the amount of video that must be sent over the network and shortens the time between an event occurring and an alert being raised.

IV. ADVANTAGES AND APPLICATIONS

Major Advantages: Real-time threat detection, reduced bandwidth usage through edge processing, lower response latency thanks to 5G, scalability across large camera networks, and reduced dependence on constant human monitoring.

Key Applications: Smart city public safety, campus and building security, traffic and license-plate monitoring, industrial and perimeter surveillance, and crowd monitoring at large public events.

V. CHALLENGES AND FUTURE SCOPE

Key challenges include protecting the privacy of individuals captured on camera, securing video and alert data as it moves across the 5G network, maintaining accuracy across varying lighting and weather conditions, and managing the compute and storage costs of running AI models on a large number of edge devices. Future work can explore federated learning so cameras can improve shared models without transmitting raw video, integration with 6G for even lower latency, and explainable AI so that automated alerts can be reviewed and trusted by human operators.

VI. CONCLUSION

AI-enabled smart surveillance over 5G networks offers a practical path toward faster, more scalable, and more automated monitoring than traditional CCTV systems. By processing video at the edge and using 5G for fast, coordinated communication, these systems can detect and respond to events in near real time while reducing bandwidth and manual review effort. Continued attention to privacy, security, and robustness will be essential for responsible deployment in public and private settings.

REFERENCES

- [1] S. Kumar et al., "A Survey on AI-Based Video Surveillance Systems: Techniques and Challenges," IEEE Access.
- [2] M. Ali et al., "Deep Learning for Real-Time Object Detection in Smart Surveillance," IEEE Transactions on Circuits and Systems for Video Technology, 2022.
- [3] "5G-Enabled Edge Intelligence for Smart City Video Analytics," Journal of Network and Computer Applications, 2023.
- [4] "AI-Driven Anomaly Detection for Smart Surveillance Networks," Future Generation Computer Systems, 2024.

RESEARCH ARTICLE

AI-Enabled Smart Surveillance Systems over 5G Networks

A Detailed Study of AI-Driven Video Analytics and 5G-Enabled Real-Time Monitoring

ABSTRACT Video surveillance has grown from a handful of fixed CCTV cameras to large, distributed networks spanning cities and campuses, making manual monitoring impractical at scale. This paper presents a detailed study of Artificial Intelligence (AI) methods for smart surveillance operating over 5G networks, covering object detection, motion tracking, facial and license-plate recognition, and anomaly detection. It analyzes how edge-based AI inference combined with 5G's low latency and high bandwidth enables near-real-time event detection while limiting the volume of raw video that must be transmitted. The study reviews evaluation criteria such as detection accuracy, response latency, and bandwidth savings, and examines open challenges around privacy, data security, and robustness of AI models under varying real-world camera conditions.

INDEX TERMS Artificial Intelligence, 5G, Smart Surveillance, Object Detection, Edge Computing, Video Analytics, Anomaly Detection.

I. INTRODUCTION

Video surveillance infrastructure has expanded rapidly across cities, transportation hubs, and campuses, but the value of that footage is limited by how quickly and accurately it can be analyzed. Traditional systems depend on human operators watching multiple screens, which does not scale and is prone to missed events. The combination of Artificial Intelligence (AI) for automated video understanding and 5G networks for fast, reliable connectivity creates an opportunity to build surveillance systems that detect and respond to events automatically, with minimal delay between capture and alert.

II. RELATED APPROACHES

Surveillance analytics approaches fall broadly into three categories. Rule-based systems (e.g., simple motion or trip-wire detection) are computationally cheap but generate frequent false alarms and cannot distinguish between benign and genuinely suspicious activity. Cloud-centric AI approaches send all video to centralized servers for deep-learning analysis, which is accurate but introduces latency and consumes significant bandwidth as camera counts grow. Edge-AI approaches run detection models directly on or near the camera, trading some model complexity for much lower latency and bandwidth use — an approach that becomes far more effective when paired with 5G connectivity between edge nodes and the control center.

III. AI TECHNIQUES FOR SURVEILLANCE

Convolutional neural networks and single-shot object detectors identify people, vehicles, and objects of interest in each frame. Multi-object tracking algorithms link detections across frames to follow movement paths and flag loitering or unauthorized entry into restricted zones. Facial and license-plate recognition

modules support identity verification at checkpoints, while autoencoder- or clustering-based anomaly detection flags activity that deviates from a learned baseline, such as sudden crowd formation or an abandoned object. 5G's low latency allows these edge-level detections to be corroborated across multiple camera feeds in near real time, improving accuracy over any single camera acting alone.

IV. EVALUATION METRICS

Common Metrics: Detection and recognition accuracy, false-alarm rate, end-to-end alert latency, bandwidth consumption per camera, and system throughput under a large number of simultaneous video streams.

Trade-offs: Higher-accuracy models are typically larger and slower, which conflicts with the low-latency goals of real-time surveillance, so deployments often use lighter models at the edge and reserve heavier models for follow-up verification in the cloud.

V. OPEN CHALLENGES

Deploying AI-enabled surveillance at scale raises challenges beyond raw detection accuracy: protecting the privacy of individuals recorded on camera, securing video and metadata as it travels over the 5G network, maintaining detection accuracy across lighting, weather, and camera-angle variation, and managing the cost of running AI inference across a large number of distributed edge devices. Regulatory and ethical constraints on facial recognition and biometric data add a further layer of complexity that purely technical solutions cannot resolve on their own.

VI. CONCLUSION

This study reviewed how AI techniques — object detection, tracking, recognition, and anomaly detection — combine with 5G connectivity to enable smart surveillance systems that respond to events in near real time. While edge-AI processing over 5G offers clear gains in latency and bandwidth efficiency compared to cloud-centric designs, practical deployment still requires careful attention to privacy, security, and robustness. These findings support continued research into efficient, privacy-aware AI models suited to real-world surveillance deployments.

REFERENCES

- [1] W. Liu et al., "SSD: Single Shot MultiBox Detector," in Proc. European Conference on Computer Vision (ECCV), 2016.
- [2] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," arXiv preprint, 2018.
- [3] "Edge AI for Real-Time Video Analytics over 5G Networks: A Survey," IEEE Communications Surveys & Tutorials, 2022.
- [4] "Deep Learning-Based Anomaly Detection for Smart Video Surveillance Systems," IEEE Access, 2023.